

7.THULIUM FIBER LASER SYSTEM

TECHNICAL SPECIFICATION

60 W THULIUM FIBER LASER SYSTEM FOR UROLOGY

1. General Specification

Sl. No.	Parameter	Minimum Tender Requirement
1	Equipment	Thulium Fiber Laser (TFL) Surgical System
2	Application	Urology, Endourology, Lithotripsy and Soft-Tissue Surgery
3	Laser Technology	Thulium Fiber Laser
4	Maximum Average Output Power	Minimum 60 W
5	Wavelength	Approximately 1.9 µm
6	Emission Modes	Continuous and Pulsed
7	Maximum Frequency	Up to 2500 Hz or better
8	Pulse Energy	Adjustable up to 6 J or better
9	Pulse Duration	Multiple/adjustable pulse duration settings
10	Beam Delivery	Optical fibre delivery
11	Fibre Compatibility	Disposable and reusable fibres
12	Fibre Diameter	Compatible with a wide range of fibre diameters, including small-diameter fibres
13	Aiming Beam	Integrated visible aiming beam
14	Cooling	Air-cooled system
15	User Interface	Touchscreen-based control interface
16	Footswitch	Double/multifunction footswitch
17	Operating Modes	Suitable for lithotripsy and soft-tissue procedures
18	Mobility	Compact mobile system with trolley/cart
19	Power Supply	Suitable for standard hospital electrical outlet, subject to manufacturer's specifications

2. LASER PERFORMANCE

The system shall provide:

- Maximum average power of at least **60 W**.
- Adjustable pulse energy up to **6 J** or better.
- Adjustable repetition rate/frequency up to **2500 Hz** or better.
- Continuous-wave emission.
- Pulsed emission.
- Super-pulsed operation, where supported.
- Multiple pulse-duration settings.
- Stable laser output during prolonged procedures.
- Precise soft-tissue cutting and ablation.
- Efficient stone fragmentation.
- Low retropulsion capability.
- Coagulation and haemostasis capability.

3. UROLOGICAL APPLICATIONS

The system shall be suitable for, as applicable and within regulatory clearance:

Stone Management

- Renal calculi
- Ureteric calculi
- Bladder calculi
- Endoscopic lithotripsy
- Dusting
- Fragmentation
- Popcorning
- Fine stone dusting
- Retropulsion-controlled lithotripsy

Prostate Surgery

- Laser enucleation of prostate
- Prostate ablation
- Prostate resection
- BPH treatment

Soft-Tissue Procedures

- Incision
- Excision
- Resection
- Ablation
- Coagulation
- Haemostasis
- Urethral stricture incision
- Bladder neck incision
- Treatment of appropriate urinary tract soft-tissue lesions

4. FIBRE DELIVERY SYSTEM

The system shall support:

1. Front-firing optical fibres.
2. Small-diameter fibres suitable for flexible ureteroscopy.
3. Fibres suitable for rigid ureteroscopy.
4. Disposable fibres.
5. Reusable fibres, where applicable.
6. Fibre sizes suitable for lithotripsy.
7. Fibre sizes suitable for soft-tissue surgery.
8. Fibre compatibility with standard endoscopic accessories.
9. Secure fibre connection.
10. Easy fibre replacement.
11. Fibre identification and safety monitoring, where available.

Tender recommendation: The bidder shall supply an initial quantity of appropriate fibres based on the hospital's expected **URS/RIRS, lithotripsy** workload.

5. CONTROL & USER INTERFACE

The system shall have:

- Colour touchscreen display.
- User-friendly graphical interface.
- Selection of continuous/pulsed mode.
- Power adjustment.
- Pulse energy adjustment.
- Frequency adjustment.
- Pulse duration adjustment.
- Ready/standby indication.
- Laser emission indication.
- Error/fault display.
- Treatment parameter display.
- User-selectable treatment presets.
- Ability to save frequently used treatment settings, where available.

6. FOOTSWITCH

The system shall be supplied with a **double/multifunction footswitch** providing:

- Laser activation.
- Control of two selectable treatment modes.

- Ready/standby control, where applicable.
- Hands-free operation.
- Safe activation mechanism.
- Robust design suitable for OT use.
- Long connecting cable.

7. COOLING SYSTEM

The laser shall have:

- Integrated air-cooling system.
- Air exhaust system.
- Thermal protection.
- Automatic shutdown in case of overheating.
- Continuous temperature monitoring, where applicable.
- No requirement for external water-cooling infrastructure.

8. LASER SAFETY

The system shall include:

- Key-operated activation.
- Emergency stop button.
- Standby and ready modes.
- Audible laser emission indication.
- Visual laser emission indication.
- Footswitch safety mechanism.
- Door interlock provision, where required.
- Laser warning signage.
- Appropriate laser safety goggles.
- Over-temperature protection.
- System fault monitoring.
- Automatic shutdown in case of critical fault.
- Appropriate laser safety compliance.

9. REQUIRED ACCESSORIES

The complete system shall be supplied with:

Item	Quantity
60 W Thulium Fiber Laser Console	1
Dedicatd Trolley/Cart	1
Double/Multifunction Footswitch	1

Item	Quantity
Laser Safety Goggles	Minimum 2 pairs
Laser Warning Sign	1 set
Power Cable	1
Fibre Connection Accessories	1 set
Front-Firing Fibres	As per hospital requirement
Small-Diameter Lithotripsy Fibres	As per hospital requirement
Soft-Tissue	As per hospital requirement
Disposable Fibres	Initial supply
Fibre Stripper/Cleaver	1 each, if applicable
User Manual	1
Service Manual	1
Installation & Commissioning	Included
User Training	Included
Biomedical Engineer Training	Included

10. ELECTRICAL REQUIREMENTS

The bidder shall specify:

- Input voltage.
- Frequency.
- Maximum power consumption.
- Maximum current requirement.
- Recommended electrical protection.
- Earthing requirement.
- UPS requirement, if any.
- Standard hospital socket compatibility.

11. WARRANTY & SERVICE

Parameter	Minimum Tender Requirement
Comprehensive Warranty	Minimum 3 years from installation/commissioning
Laser Source	Covered under warranty
Console	Covered under warranty
Footswitch	Covered under warranty
Cooling System	Covered under warranty
Service Response	Within 24–48 hours

Parameter	Minimum Tender Requirement
Preventive Maintenance	Minimum 3 visits/year during warranty
Calibration	As per manufacturer's recommended schedule
Spare Parts	Availability for minimum 7–10 years
Service Support	Authorized service support in India
Training	Urologist, OT staff and Biomedical Engineer
Installation	Complete installation and commissioning
Demonstration	Pre-bid/demo if requested

12.Warranty & After-Sales Support

- **Comprehensive Warranty:** Minimum 3 years from the date of installation/commissioning.
- **Preventive Maintenance:** Minimum 4 preventive maintenance visits per year during the warranty period.
- **Spare Parts Support:** Availability and support for genuine spare parts for a minimum period of 10 years.
- **Backup Machine:** In the event of any equipment complaint or breakdown, a suitable backup machine shall be arranged within 12 hours until the original equipment is restored to operational condition.
- **AMC/CAMC Charges:** The AMC/CAMC rate as a percentage of the equipment purchase value, applicable after the warranty period, shall be finalized and frozen at the time of purchase and clearly mentioned in the Purchase Order.
- **Manufacturer Authorization & Warranty:** All authorization letters, warranty documents, and related commitments shall be issued directly by the original manufacturer/company, and not solely by the vendor/dealer.